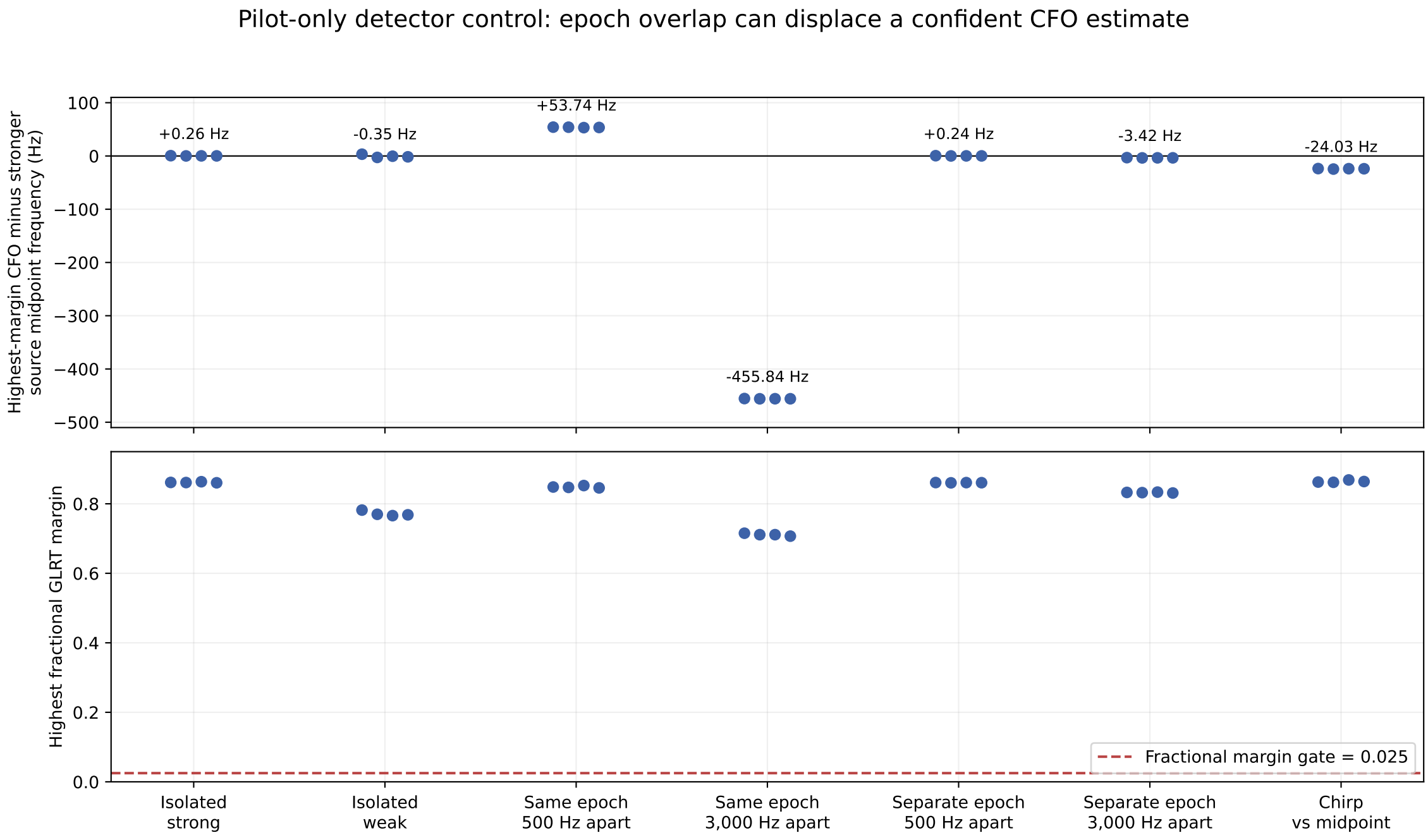
Pilot-only detector control: epoch overlap can displace a confident CFO estimate
Highest-margin CFO minus stronger source midpoint frequency (Hz)
100
0
−100
−200
−300
−400
−500
+0.26 Hz
-0.35 Hz
+53.74 Hz
-455.84 Hz
+0.24 Hz
-3.42 Hz
-24.03 Hz
Highest fractional GLRT margin
0.8
0.6
0.4
0.2
0.0
Fractional margin gate = 0.025
Isolated strong
Isolated weak
Same epoch 500 Hz apart
Same epoch 3,000 Hz apart
Separate epoch 500 Hz apart
Separate epoch 3,000 Hz apart
Chirp vs midpoint
Four outputs per case: two seeds × two receivers. Noise-only: 0/4 accepted. Known template boundary mismatch; no field or position claim.